

GhostObjects: Instructing Robots by Manipulating Spatially Aligned Virtual Twins in Augmented Reality

Lauren W. Wang
Computer Science
Princeton University
Princeton, New Jersey, USA
wlauren@princeton.edu

Parastoo Abtahi
Computer Science
Princeton University
Princeton, New Jersey, USA
parastoo@princeton.edu

Abstract

Robots are increasingly capable of autonomous operations, yet human interaction remains essential for issuing personalized instructions. Instead of directly controlling robots through Programming by Demonstration (PbD) or teleoperation, we propose giving instructions by interacting with *GhostObjects*—world-aligned, life-size virtual twins of physical objects—in augmented reality (AR). By direct manipulation of *GhostObjects*, users can precisely specify physical goals and spatial parameters, with features including real-world lasso selection of multiple objects and snapping back to default positions, enabling tasks beyond simple pick-and-place.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Mixed / augmented reality**; *Interaction techniques*.

Keywords

Human-robot interaction, collaborative tasks, robotics, robot instructions, end-user robot programming, mixed reality, AR

ACM Reference Format:

Lauren W. Wang and Parastoo Abtahi. 2025. GhostObjects: Instructing Robots by Manipulating Spatially Aligned Virtual Twins in Augmented Reality. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758451>

1 Introduction

We are increasingly surrounded by autonomous robots that augment our capabilities in a variety of applications, including household cleaning, package delivery, and home furnishing. These contexts require users to give explicit instructions on specific objects at a precise location in space (e.g., where to put which clothes, which shelf to load the unboxed products, where to hang the painting).

Augmented reality (AR) has emerged as a promising approach for human-robot interaction that requires spatial manipulation, including end-user robot programming that modifies waypoints or trajectories for robot motion planning [3, 5, 6, 10, 11, 13], improving

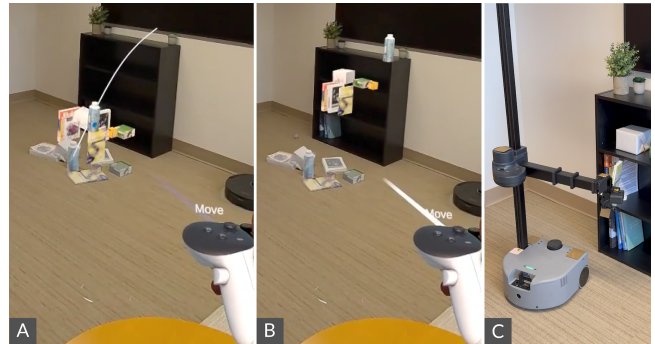


Figure 1: Moving a pile of *GhostObjects* by (A) directly manipulating the world-aligned, life-size virtual twins of the real objects and (B) snapping them onto the shelf by releasing along the default trajectory. (C) A robot receives the instruction and carries out the task.

teleoperation of robotic arms and drones through AR visualizations [4, 8], and communicating robot intent [14, 15, 17].

Prior work has explored a wide range of methods for instructing robots. GhostAR [2] uses demonstrative role-playing, an embodied authoring approach that requires a high user understanding of robot capabilities and demonstration costs. Rather than relying on captured human motion, SayCan [1] employs a textual conversational agent equipped with high-level semantic knowledge to perform generalized tasks. Yet natural language inputs are known to suffer from issues of discoverability and ambiguity. Marcer [10] also uses language commands, but introduces AR to visualize virtual twins of robots and objects for clarity. Still, these AR visualizations aren't interactive, so the issues with language inputs remain. ImageInThat [12] reduces this friction by enabling direct manipulation of pixels though dragging corresponding objects in an image editor. However, it remains limited to a 2D interface. HoloSpot [7] moves into the 3D world, enabling users to drag and drop a single virtual object to issue pick-and-place commands to the robot in a miniaturized world that is not co-located. Its World-in-Miniature (WiM) setup, while effective for remote teleoperation, is not ideal when robot instructions demand co-located viewing, as in scenarios like arranging items in new locations or deciding where to hang a painting for the best view. X-HR Training [16] similarly showed examples of human-robot collaboration in 3D AR, but direct manipulation of virtual twins was used for Learning by Demonstration purposes without physical counterparts, and was not spatially aligned with real objects. In other examples, co-located virtual objects served as non-interactive AR visualizations for previewing instruction outcomes.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758451>

Instead of directly controlling robots through expert authoring, our approach centers on interacting with target objects. Rather than relying on language commands, we enable direct manipulation for greater precision. To support co-located tasks and life-size manipulation, we develop a 3D, aligned AR environment. In sum, we propose instructing robots by directly interacting with virtual twins of real-world objects—*GhostObjects*—in spatially aligned physical space using augmented reality (AR). Users can raycast to select an object in passthrough view and directly manipulate the *GhostObject*. In Figure 1, to instruct a robot to move a pile of items, the user can grab the *GhostObjects* and place them freely within a physics-enabled environment or snap them back to default locations.

2 Ghost Objects

2.1 Single-object Raycast Selection

Unlike traditional interfaces (e.g. WiM [7]) that require users to mentally map targets into digital space, our system enables direct AR interaction with co-located objects. To select an object, the user initiates a raycast into the environment by pointing the controller. Pressing the trigger button selects the first object intersected by the ray.

2.2 Multi-object Lasso Selection

Additionally, users can select multiple objects by holding the trigger button and drawing a lasso to outline an arbitrary shape around their desired targets. The ray projected from the controller intersects with surfaces in the environment (Figure 2A). As the user moves the controller, intersection points connect, forming a closed boundary when the trigger is released. This lassomed boundary creates a volume extending towards the controller, and objects that overlap this volume are grouped and selected together.

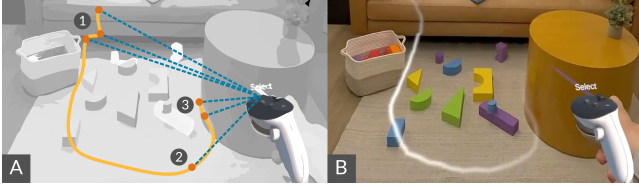


Figure 2: Lasso selection. (A) An illustration showing the lasso intersecting the sofa (1), carpet (2), and table (3). The yellow line represents the lasso, and the blue line represents the raycast. (B) The AR headset view of the lasso selection.

2.3 Spatially Aligned Virtual Twins

When an object is selected, a virtual twin—called a *GhostObject*—is immediately overlaid on the real object in AR. *GhostObjects* are life-sized and spatially aligned with the real world. We utilize Meta Quest’s Space Setup feature, which scans the scene and generates bounding boxes for spatial anchors. We then leverage Meta’s Mixed Reality Utility Kit (MRUK) to access the positions of these spatial anchors, which serve as reference points for spawning *GhostObjects* in their relative poses. This ensures that objects consistently appear in the correct physical location, eliminating repeated manual calibration. The virtual twins provide users with immediate feedback and help them visualize the anticipated outcomes of robot tasks.

2.4 Direct Manipulation of *GhostObjects*

GhostObjects can be directly manipulated, minimizing the gulf between users’ intentions and task execution [9]. Direct manipulation lets *GhostObjects* be moved, scaled, aligned, and visually modified in AR. Users can hold the trigger button to grab, move, and place *GhostObjects*, which conform to the environment’s simulated physics. Beyond relocation, users can also modify *GhostObjects* that change form or state—for example, adjusting the liquid level inside a bottle during filling tasks or compressing empty boxes for recycling.

2.5 Snap to Default

Each *GhostObject* has a customizable default location that stores the pose where the object is expected to be. When moved, an arched trajectory line appears from its current position to the default location. Releasing the *GhostObject* along this path causes it to snap back to default. Users can redefine the default location by placing the *GhostObject* elsewhere and saving that as the default.

3 Application Scenarios

3.1 Home Tidying

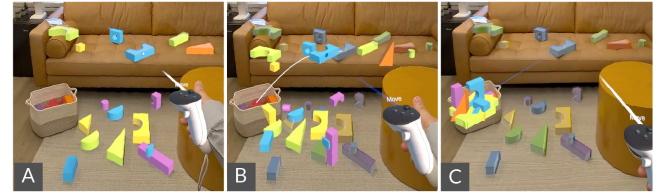


Figure 3: Relocating scattered foam blocks to a basket using direct manipulation and the snap-to-default feature.

We demonstrate a walkthrough of a domestic scenario where a user instructs a robot to move foam blocks into a basket beside the sofa. After lasso selecting the blocks (Figure 2B), she points the controller at one of the *block GhostObjects* and holds the trigger to grab it (Figure 3A). Since all the blocks are selected, they move as a group, with a trajectory line connecting them to the basket to indicate their default location (Figure 3B). She moves the blocks along this path and releases the trigger, snapping them into the basket (Figure 3C).

3.2 Filling Bottle

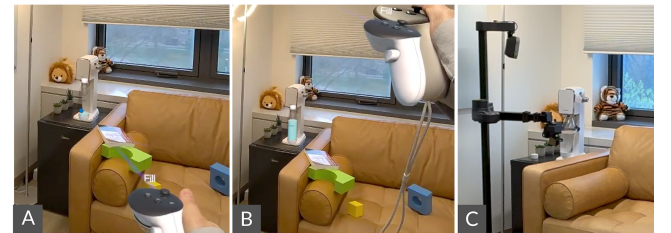


Figure 4: Filling water *GhostObject* to a desired level.

While we focus on pick-and-place actions, *GhostObjects* can also extend to robot tasks involving state change or deformation. For example, the user finds a soda machine beside the sofa and wants

the robot to fill the bottle with water. She holds the trigger and raises her hand up, deforming the *water GhostObject* to specify the desired fill level (Figure 4). Once the user's instructions are complete, the action and spatial parameters are sent to the robot.

4 Conclusion and Future Work

In this work, we present *GhostObjects*—world-aligned, life-size virtual twins of physical objects in augmented reality (AR)—that can be directly manipulated as a means of instructing robots. In the future, we plan to explore various object affordances and robot tasks to further experiment with *GhostObjects*. We also aim to leverage modern reconstruction techniques and vision-language models to automate the creation of *GhostObjects*, enhancing their generalizability. Additionally, we intend to conduct further evaluations of the efficacy of using *GhostObjects* in real-world robot tasks.

References

- [1] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Daniel Ho, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Eric Jang, Rosario Jauregui Ruano, Kyle Jeffrey, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Kuang-Huei Lee, Sergey Levine, Yao Lu, Linda Luu, Carolina Parada, Peter Pastor, Jornell Quiambao, Kanishka Rao, Jarek Rettinghouse, Diego Reyes, Pierre Sermanet, Nicolas Sievers, Clayton Tan, Alexander Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Mengyuan Yan, and Andy Zeng. 2022. Do As I Can and Not As I Say: Grounding Language in Robotic Affordances.
- [2] Yuanzhi Cao, Tianyi Wang, Xun Qian, Pawan S. Rao, Manav Wadhawan, Ke Huo, and Karthik Ramani. 2019. GhostAR: A Time-space Editor for Embodied Authoring of Human-Robot Collaborative Task with Augmented Reality. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 521–534. <https://doi.org/10.1145/3332165.3347902>
- [3] Yuanzhi Cao, Zhuangying Xu, Fan Li, Wentao Zhong, Ke Huo, and Karthik Ramani. 2019. V. ra: An in-situ visual authoring system for robot-iot task planning with augmented reality. In *Proceedings of the 2019 on designing interactive systems conference*. Association for Computing Machinery, New York, NY, USA, 1059–1070.
- [4] Jiaqi Chen, Boyang Sun, Marc Pollefeys, and Hermann Blum. 2024. A 3d mixed reality interface for human-robot teaming. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, IEEE, Japan, 11327–11333.
- [5] H. C. Fang, S. K. Ong, and A. Y. C. Nee. 2012. Interactive robot trajectory planning and simulation using Augmented Reality. *Robot. Comput.-Integr. Manuf.* 28, 2 (April 2012), 227–237. <https://doi.org/10.1016/j.rcim.2011.09.003>
- [6] Samir Yitzhak Gadre, Eric Rosen, Gary Chien, Elizabeth Phillips, Stefanie Tellex, and George Konidaris. 2019. End-User Robot Programming Using Mixed Reality. In *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, Canada, 2707–2713. <https://doi.org/10.1109/ICRA.2019.8793988>
- [7] Pablo Soler Garcia, Petar Lukovic, Lucie Reynaud, Andrea Sgobbi, Federica Bruni, Martin Brun, Marc Zünd, Riccardo Bollati, Marc Pollefeys, Hermann Blum, and Zuria Bauer. 2024. HoloSpot: Intuitive Object Manipulation via Mixed Reality Drag-and-Drop. arXiv:2410.11110 [cs.RO] <https://arxiv.org/abs/2410.11110>
- [8] Hooman Hedayati, Michael Walker, and Daniel Szafrir. 2018. Improving Collocated Robot Teleoperation with Augmented Reality. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (Chicago, IL, USA) (HRI '18). Association for Computing Machinery, New York, NY, USA, 78–86. <https://doi.org/10.1145/3171221.3171251>
- [9] Edwin L. Hutchins, James D. Hollan, and Donald A. Norman. 1985. Direct manipulation interfaces. *Hum.-Comput. Interact.* 1, 4 (Dec. 1985), 311–338. https://doi.org/10.1207/s15327051hci0104_2
- [10] Bryce Ikeda, Maitrey Gramopadhye, LillyAnn Nekervis, and Daniel Szafrir. 2025. MARCER: Multimodal Augmented Reality for Composing and Executing Robot Tasks. In *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE, Australia, 529–539.
- [11] Rasmus Skovhus Lunding, Mille Skovhus Lunding, Tiare Feuchtnet, Marianne Graves Petersen, Kaj Grønbaek, and Ryo Suzuki. 2024. RoboVisAR: Immersive Authoring of Condition-based AR Robot Visualisations. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (Boulder, CO, USA) (HRI '24). Association for Computing Machinery, New York, NY, USA, 462–471. <https://doi.org/10.1145/3610977.3634972>
- [12] Karthik Mahadevan, Blaine Lewis, Jiannan Li, Bilge Mutlu, Anthony Tang, and Tovi Grossman. 2025. ImageInThat: Manipulating Images to Convey User Instructions to Robots. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, Australia, 757–766. <https://doi.org/10.1109/HRI61500.2025.10974179>
- [13] Camilo Perez Quintero, Sarah Li, Matthew KXJ Pan, Wesley P. Chan, H.F. Machiel Van der Loos, and Elizabeth Croft. 2018. Robot Programming Through Augmented Trajectories in Augmented Reality. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, Spain, 1838–1844. <https://doi.org/10.1109/IROS.2018.8593700>
- [14] Eric Rosen, David Whitney, Elizabeth Phillips, Gary Chien, James Tompkin, George Konidaris, and Stefanie Tellex. 2019. Communicating and controlling robot arm motion intent through mixed-reality head-mounted displays. *Int. J. Rob. Res.* 38, 12–13 (Oct. 2019), 1513–1526. <https://doi.org/10.1177/0278364919842925>
- [15] Michael Walker, Hooman Hedayati, Jennifer Lee, and Daniel Szafrir. 2018. Communicating Robot Motion Intent with Augmented Reality. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (Chicago, IL, USA) (HRI '18). Association for Computing Machinery, New York, NY, USA, 316–324. <https://doi.org/10.1145/3171221.3171253>
- [16] Chao Wang, Anna Belardinelli, Stephan Hasler, Theodoros Stouraitis, Daniel Tanneberg, and Michael Gienger. 2023. Explainable Human-Robot Training and Cooperation with Augmented Reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 449, 5 pages. <https://doi.org/10.1145/3544549.3583889>
- [17] Danny Zhu. 2017. *Augmented Reality Visualization for Autonomous Robots*. Ph.D. Dissertation. Air Force Research Laboratory.